EduMate_Requirements

2.2.1: Define functional requirements and use cases

*Functional Requirements

ID	Requirement	Feature	Priority	Source
FR-01.1	The system shall allow users (students and lecturers) to register a new account using their email.	F01	Must	PA1 Vision
FR-01.2	The system shall allow users to log in securely with their credentials.	F01	Must	PA1 Vision
FR-02.1	The system shall allow students to browse a list of available courses and lecturers.	F02	Must	Stakeholder Interview
FR-02.2	The system shall allow students to enroll in a selected course.	F02	Must	Stakeholder Interview
FR-03.1	The system shall automatically generate a combined weekly timetable for a student based on their enrolled courses.	F03	Must	PA1 Vision
FR-04.1	The system shall allow lecturers to create, update, and delete assignment and exam deadlines for their courses.	F04	Must	Stakeholder Interview
FR-04.2	The system shall display upcoming deadlines and tasks on the student's dashboard.	F04	Must	Stakeholder Interview
FR-05.1	The system shall allow students to post text-based questions within a specific enrolled course.	F05	Must	PA1 Vision

FR-05.2	The system shall allow lecturers to reply to student questions within their assigned courses.	F05	Must	PA1 Vision
FR-06.1	The system shall send push notifications to students 24 hours before an upcoming deadline.	F06	Should	Team Discussion
FR-07.1	The system shall display a visual chart showing completed vs. pending tasks for students.	F07	Should	Team Discussion

*Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Register and Log in	Student, Lecturer	Must	FR-01.1, FR-01.2
UC-02	Browse and Enroll in Courses	Student	Must	FR-02.1, FR-02.2
UC-03	View Automated Timetable	Student	Must	FR-03.1
UC-04	Manage Course Deadlines	Lecturer	Must	FR-04.1
UC-05	Track Upcoming Deadlines	Student	Must	FR-04.2
UC-06	Ask a Question	Student	Must	FR-05.1
UC-07	Answer a Question	Lecturer	Must	FR-05.2

2.2.2: Define business rules and non-functional requirements

*Non-Functional Requirements

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	Under normal load (100 concurrent users), the	Using performance testing tools to simulate

		system shall load the user's timetable and dashboard in ≤ 2 seconds.	users and measure load time.
NFR-02	Security	User passwords shall be encrypted using a strong hashing algorithm before being stored in the database.	Code review of the authentication module and database inspection to verify hashed strings.
NFR-03	Usability	The web application shall be fully responsive and display correctly on both desktop monitors and mobile screens.	Manual testing using browser developer tools and real mobile devices across different screen resolutions.
NFR-04	Reliability	The system shall maintain an uptime of 99% during the main academic semester (excluding scheduled maintenance).	Server monitoring logs and uptime tracking services.

*Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	A student can only view the Q&A section and post questions in courses they are actively enrolled in.	Academic Integrity / System Policy
BR-02	Only the lecturer formally assigned to a course is authorized to create or modify its deadlines, schedules, and reply as the official instructor.	Academic Management Policy
BR-03	Users must be fully authenticated (logged in) to access any scheduling, course, or Q&A data. Anonymous access is strictly prohibited.	Security Standard